Solar Ca appendix

Sun · Ca · downloadable appendix

Document	Solar Ca appendix
Subject	Solar photospheric abundance appendix · Ca
Author	Ryan Schmitt
Date	2026-11-07
Publisher	The Exoplanet Codex · exoplanetcodex.org
Version	feed v1.345
Source	science commit 798869548bcf
Web	https://exoplanetcodex.org/systems/sol/elements/ca/

Solar calcium: 6.405 ± 0.049

Asplund, Amarsi & Grevesse 2021: 6.30 ± 0.03 · this measurement +0.105 dex

The Ca I value sits about 0.1 dex above Asplund et al. 2021. They fit the light from the centre of the solar disc; we fit the light of the whole disc, as we must for other stars. Fitting the disc centre with the same lines and code gives their value, so the difference comes from the strong lines' edge-of-disc light, not from the line data. The Ca II lines agree with Asplund et al. 2021. The infrared values read about 0.1 dex higher still. On iron lines that also have laboratory values, the Cepheid-calibrated oscillator strengths sit about 0.14 dex low, which would push a calcium abundance up by about that much; the infrared products are shown as measured, with that caveat.

Asplund, Amarsi & Grevesse (2021) give no line recipe we can repeat for this element, so this is our best HARPS product: the Reference Grade analysis with the most complete treatment, then the most lines.

VIS: 6.405 ± 0.049 · HARPS · VIS · Ca I · Asplund et al. 2021 lines · 3D NLTE (Amarsi)

Highlighted Products

VIS

6.413 ± 0.048

Ca I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · Kitt Peak (1984, molecfit) · Reference Grade

red-optical

6.466 ± 0.059

Ca II · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · Kitt Peak (1984, molecfit) · Reference Grade

NIR

6.500 ± 0.058

Ca I · Codex-graded lines · <3D> LTE (Gerber) · Kitt Peak (1984, molecfit) · Codex Grade

Line grades

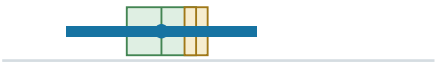
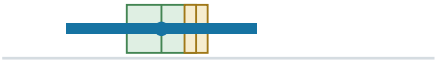
Every product is measured on one of three grades of spectral lines and the grades are never mixed in one number.

Grade	What it means	This element
Reference Grade	lines from a published solar line set (Asplund et al. 2021 and similar), measured with the oscillator strengths the authors used	99 products -- Ca I · Asplund et al. 2021 lines; Ca I · reference lines; Ca II · Asplund et al. 2021 lines

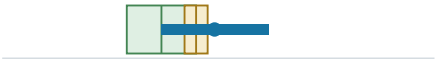
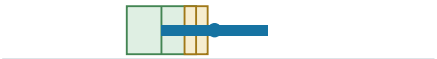
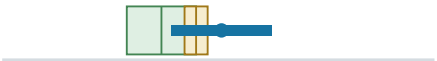
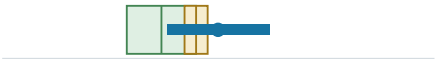
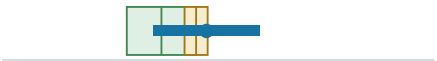
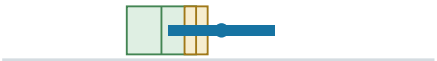
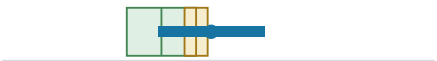
Grade	What it means	This element
Codex Grade	lines chosen by the Codex whose oscillator strength carries a published uncertainty (laboratory measurement, NIST accuracy class A-C, or the paper's own stated error), between 5% and 60% deep	89 products -- Ca I · Codex-graded lines; Ca II · Codex-graded lines
Deep Grade	the same oscillator-strength rule on lines deeper than 60% -- stronger lines, more sensitive to damping and the model atmosphere	none for the Sun: every line in this grade is excluded as saturated or an outlier (see Excluded lines)

Error-bar Forest Plot

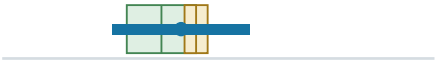
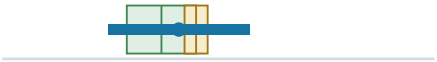
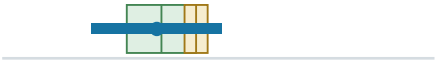
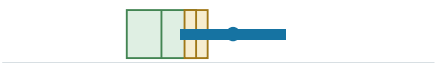
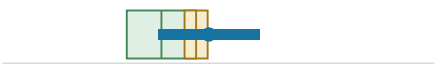
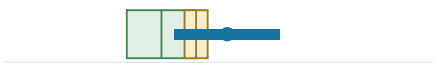
VIS — HARPS
A(Ca) dex; axis 6.16–6.54

Ca II · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines		6.293 ±0.066
Ca II · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 2 lines		6.300 ±0.083
Ca II · Asplund et al. 2021 lines · 1D NLTE (Bergemann) Reference Grade · 2 lines		6.290 ±0.068
Ca II · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 2 lines		6.258 ±0.066
Ca II · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 2 lines		6.271 ±0.063
Ca II · Asplund et al. 2021 lines · <3D> LTE (Gerber) Reference Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		6.349 ±0.062
Ca II · Asplund et al. 2021 lines · <3D> NLTE (Gerber) Reference Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		6.332 ±0.065
Ca II · Codex-graded lines · 1D LTE Codex Grade · 2 lines		6.293 ±0.066
Ca II · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 2 lines		6.300 ±0.083
Ca II · Codex-graded lines · 1D NLTE (Bergemann) Codex Grade · 2 lines		6.290 ±0.068
Ca II · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 2 lines		6.258 ±0.066
Ca II · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 2 lines		6.271 ±0.063
Ca II · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		6.349 ±0.062
Ca II · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		6.332 ±0.065
Ca I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 10 lines		6.343 ±0.046
Ca I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 10 lines		6.405 ±0.049
Ca I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) Reference Grade · 10 lines		6.322 ±0.045

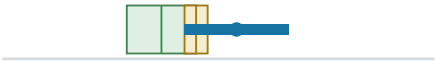
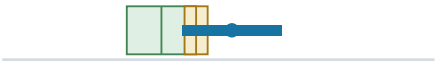
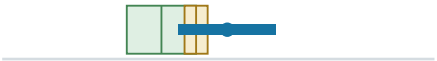
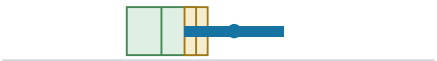
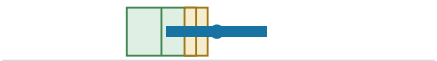
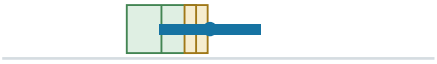
VIS — HARPS
A(Ca) dex; axis 6.16–6.54

Ca I · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 10 lines		6.346 ±0.046
Ca I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 10 lines		6.346 ±0.046
Ca I · Asplund et al. 2021 lines · <3D> LTE (Gerber) Reference Grade · 10 lines · EXPERIMENTAL-NOT-ADOPTED		6.438 ±0.049
Ca I · Asplund et al. 2021 lines · <3D> NLTE (Gerber) Reference Grade · 10 lines · EXPERIMENTAL-NOT-ADOPTED		6.434 ±0.048
Ca I · Codex-graded lines · 1D LTE Codex Grade · 17 lines		6.347 ±0.045
Ca I · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 10 lines		6.405 ±0.049
Ca I · Codex-graded lines · 1D NLTE (Bergemann) Codex Grade · 17 lines		6.352 ±0.043
Ca I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 17 lines		6.336 ±0.044
Ca I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 17 lines		6.349 ±0.044
Ca I · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 17 lines · EXPERIMENTAL-NOT-ADOPTED		6.438 ±0.046
Ca I · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 17 lines · EXPERIMENTAL-NOT-ADOPTED		6.409 ±0.044
Ca I · reference lines · 1D LTE Reference Grade · 45 lines		6.339 ±0.046
Ca I · reference lines · 3D NLTE (Amarsi) Reference Grade · 10 lines		6.405 ±0.049
Ca I · reference lines · 1D NLTE (Bergemann) Reference Grade · 45 lines		6.352 ±0.046
Ca I · reference lines · 1D NLTE (Gerber) Reference Grade · 45 lines		6.310 ±0.046
Ca I · reference lines · 1D LTE (Gerber) Reference Grade · 45 lines		6.343 ±0.046
Ca I · reference lines · <3D> LTE (Gerber) Reference Grade · 45 lines · EXPERIMENTAL-NOT-ADOPTED		6.448 ±0.046
Ca I · reference lines · <3D> NLTE (Gerber) Reference Grade · 45 lines · EXPERIMENTAL-NOT-ADOPTED		6.411 ±0.046

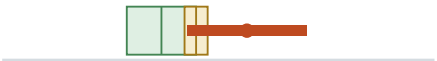
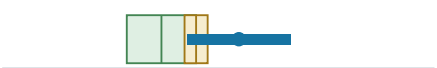
VIS — Kitt Peak (Kurucz 2005)
A(Ca) dex; axis 6.16–6.54

Ca II · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines		6.317 ±0.059
Ca II · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 2 lines		6.325 ±0.075
Ca II · Asplund et al. 2021 lines · 1D NLTE (Bergemann) Reference Grade · 2 lines		6.315 ±0.061
Ca II · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 2 lines		6.263 ±0.047
Ca II · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 2 lines		6.296 ±0.056
Ca II · Asplund et al. 2021 lines · <3D> LTE (Gerber) Reference Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		6.373 ±0.056
Ca II · Asplund et al. 2021 lines · <3D> NLTE (Gerber) Reference Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		6.351 ±0.058
Ca II · Codex-graded lines · 1D LTE Codex Grade · 2 lines		6.317 ±0.059
Ca II · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 2 lines		6.325 ±0.075
Ca II · Codex-graded lines · 1D NLTE (Bergemann) Codex Grade · 2 lines		6.315 ±0.061
Ca II · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 2 lines		6.263 ±0.047
Ca II · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 2 lines		6.296 ±0.056
Ca II · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		6.373 ±0.056
Ca II · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		6.351 ±0.058
Ca I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 10 lines		6.362 ±0.046
Ca I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 10 lines		6.412 ±0.048
Ca I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) Reference Grade · 10 lines		6.341 ±0.044
Ca I · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 10 lines		6.357 ±0.046

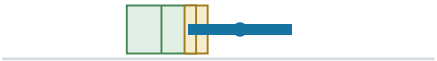
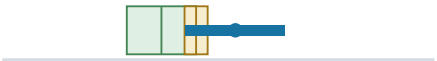
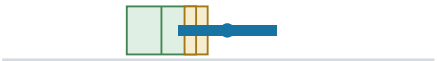
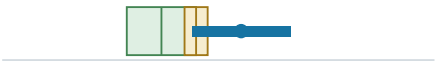
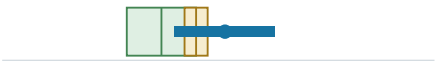
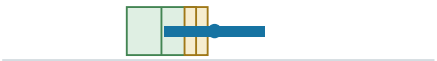
VIS — Kitt Peak (Kurucz 2005)
A(Ca) dex; axis 6.16–6.54

Ca I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 10 lines		6.365 ±0.045
Ca I · Asplund et al. 2021 lines · <3D> LTE (Gerber) Reference Grade · 10 lines · EXPERIMENTAL-NOT-ADOPTED		6.455 ±0.048
Ca I · Asplund et al. 2021 lines · <3D> NLTE (Gerber) Reference Grade · 10 lines · EXPERIMENTAL-NOT-ADOPTED		6.441 ±0.048
Ca I · Codex-graded lines · 1D LTE Codex Grade · 17 lines		6.361 ±0.043
Ca I · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 10 lines		6.412 ±0.048
Ca I · Codex-graded lines · 1D NLTE (Bergemann) Codex Grade · 17 lines		6.357 ±0.042
Ca I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 17 lines		6.341 ±0.043
Ca I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 17 lines		6.363 ±0.043
Ca I · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 17 lines · EXPERIMENTAL-NOT-ADOPTED		6.449 ±0.044
Ca I · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 17 lines · EXPERIMENTAL-NOT-ADOPTED		6.436 ±0.044
Ca I · reference lines · 1D LTE Reference Grade · 45 lines		6.337 ±0.044
Ca I · reference lines · 3D NLTE (Amarsi) Reference Grade · 10 lines		6.412 ±0.048
Ca I · reference lines · 1D NLTE (Bergemann) Reference Grade · 45 lines		6.348 ±0.044
Ca I · reference lines · 1D NLTE (Gerber) Reference Grade · 45 lines		6.298 ±0.044
Ca I · reference lines · 1D LTE (Gerber) Reference Grade · 45 lines		6.342 ±0.044
Ca I · reference lines · <3D> LTE (Gerber) Reference Grade · 45 lines · EXPERIMENTAL-NOT-ADOPTED		6.449 ±0.045
Ca I · reference lines · <3D> NLTE (Gerber) Reference Grade · 45 lines · EXPERIMENTAL-NOT-ADOPTED		6.402 ±0.044

VIS — Kitt Peak (1984, molecfit)
A(Ca) dex; axis 6.16–6.54

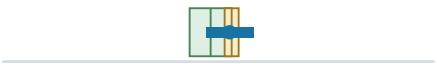
Ca II · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines		6.319 ±0.054
Ca II · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 2 lines		6.327 ±0.069
Ca II · Asplund et al. 2021 lines · 1D NLTE (Bergemann) Reference Grade · 2 lines		6.317 ±0.056
Ca II · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 2 lines		6.287 ±0.056
Ca II · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 2 lines		6.298 ±0.052
Ca II · Asplund et al. 2021 lines · <3D> LTE (Gerber) Reference Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		6.374 ±0.052
Ca II · Asplund et al. 2021 lines · <3D> NLTE (Gerber) Reference Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		6.354 ±0.053
Ca II · Codex-graded lines · 1D LTE Codex Grade · 2 lines		6.319 ±0.054
Ca II · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 2 lines		6.327 ±0.069
Ca II · Codex-graded lines · 1D NLTE (Bergemann) Codex Grade · 2 lines		6.317 ±0.056
Ca II · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 2 lines		6.287 ±0.056
Ca II · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 2 lines		6.298 ±0.052
Ca II · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		6.374 ±0.052
Ca II · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		6.354 ±0.053
Ca I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 10 lines		6.364 ±0.045
Ca I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 10 lines		6.413 ±0.048
Ca I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) Reference Grade · 10 lines		6.345 ±0.043
Ca I · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 10 lines		6.367 ±0.045

VIS — Kitt Peak (1984, molecfit)
A(Ca) dex; axis 6.16–6.54

Ca I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 10 lines		6.368 ±0.045
Ca I · Asplund et al. 2021 lines · <3D> LTE (Gerber) Reference Grade · 10 lines · EXPERIMENTAL-NOT-ADOPTED		6.459 ±0.047
Ca I · Asplund et al. 2021 lines · <3D> NLTE (Gerber) Reference Grade · 10 lines · EXPERIMENTAL-NOT-ADOPTED		6.450 ±0.047
Ca I · Codex-graded lines · 1D LTE Codex Grade · 17 lines		6.364 ±0.043
Ca I · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 10 lines		6.413 ±0.048
Ca I · Codex-graded lines · 1D NLTE (Bergemann) Codex Grade · 17 lines		6.359 ±0.042
Ca I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 17 lines		6.357 ±0.042
Ca I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 17 lines		6.369 ±0.043
Ca I · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 17 lines · EXPERIMENTAL-NOT-ADOPTED		6.460 ±0.044
Ca I · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 17 lines · EXPERIMENTAL-NOT-ADOPTED		6.444 ±0.044
Ca I · reference lines · 1D LTE Reference Grade · 45 lines		6.341 ±0.044
Ca I · reference lines · 3D NLTE (Amarsi) Reference Grade · 10 lines		6.413 ±0.048
Ca I · reference lines · 1D NLTE (Bergemann) Reference Grade · 45 lines		6.355 ±0.043
Ca I · reference lines · 1D NLTE (Gerber) Reference Grade · 45 lines		6.310 ±0.043
Ca I · reference lines · 1D LTE (Gerber) Reference Grade · 45 lines		6.346 ±0.044
Ca I · reference lines · <3D> LTE (Gerber) Reference Grade · 45 lines · EXPERIMENTAL-NOT-ADOPTED		6.462 ±0.044
Ca I · reference lines · <3D> NLTE (Gerber) Reference Grade · 45 lines · EXPERIMENTAL-NOT-ADOPTED		6.420 ±0.044

VIS axis: 6.16 · 6.26 · 6.35 · 6.44 · 6.54 A(Ca) dex

red-optical — IAG FTS
A(Ca) dex; axis 6.00–6.62

Ca II · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines		6.458 ±0.067
Ca II · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 2 lines		6.472 ±0.075
Ca II · Asplund et al. 2021 lines · 1D NLTE (Bergemann) Reference Grade · 2 lines		6.422 ±0.054
Ca II · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 2 lines		6.320 ±0.034
Ca II · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 2 lines		6.430 ±0.063
Ca II · Asplund et al. 2021 lines · <3D> LTE (Gerber) Reference Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		6.485 ±0.066
Ca II · Asplund et al. 2021 lines · <3D> NLTE (Gerber) Reference Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		6.362 ±0.034
Ca II · Codex-graded lines · 1D LTE Codex Grade · 3 lines		6.494 ±0.049
Ca II · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 2 lines		6.472 ±0.075
Ca II · Codex-graded lines · 1D NLTE (Bergemann) Codex Grade · 3 lines		6.379 ±0.046
Ca II · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 3 lines		6.327 ±0.034
Ca II · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 3 lines		6.464 ±0.047
Ca II · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 3 lines · EXPERIMENTAL-NOT-ADOPTED		6.523 ±0.048
Ca II · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 3 lines · EXPERIMENTAL-NOT-ADOPTED		6.368 ±0.034
Ca I · Codex-graded lines · 1D LTE Codex Grade · 3 lines		6.170 ±0.088
Ca I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 3 lines		6.153 ±0.085
Ca I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 3 lines		6.173 ±0.087
Ca I · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 3 lines · EXPERIMENTAL-NOT-ADOPTED		6.258 ±0.090

red-optical — IAG FTS
A(Ca) dex; axis 6.00–6.62

Ca I · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 3 lines · EXPERIMENTAL-NOT-ADOPTED		6.229 ±0.084
Ca I · reference lines · 1D LTE Reference Grade · 3 lines		6.170 ±0.088
Ca I · reference lines · 1D NLTE (Gerber) Reference Grade · 3 lines		6.153 ±0.085
Ca I · reference lines · 1D LTE (Gerber) Reference Grade · 3 lines		6.173 ±0.087
Ca I · reference lines · <3D> LTE (Gerber) Reference Grade · 3 lines · EXPERIMENTAL-NOT-ADOPTED		6.258 ±0.090
Ca I · reference lines · <3D> NLTE (Gerber) Reference Grade · 3 lines · EXPERIMENTAL-NOT-ADOPTED		6.229 ±0.084

red-optical — Kitt Peak (Kurucz 2005)
A(Ca) dex; axis 6.00–6.62

Ca II · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines		6.477 ±0.060
Ca II · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 2 lines		6.491 ±0.068
Ca II · Asplund et al. 2021 lines · 1D NLTE (Bergemann) Reference Grade · 2 lines		6.441 ±0.048
Ca II · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 2 lines		6.335 ±0.037
Ca II · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 2 lines		6.448 ±0.056
Ca II · Asplund et al. 2021 lines · <3D> LTE (Gerber) Reference Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		6.502 ±0.059
Ca II · Asplund et al. 2021 lines · <3D> NLTE (Gerber) Reference Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		6.377 ±0.037
Ca II · Codex-graded lines · 1D LTE Codex Grade · 3 lines		6.514 ±0.046
Ca II · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 2 lines		6.491 ±0.068
Ca II · Codex-graded lines · 1D NLTE (Bergemann) Codex Grade · 3 lines		6.406 ±0.043

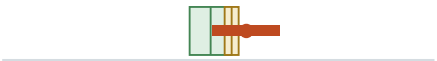
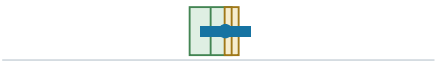
red-optical — Kitt Peak (Kurucz 2005)
A(Ca) dex; axis 6.00–6.62

Ca II · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 3 lines		6.350 ±0.036
Ca II · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 3 lines		6.485 ±0.044
Ca II · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 3 lines · EXPERIMENTAL-NOT-ADOPTED		6.544 ±0.046
Ca II · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 3 lines · EXPERIMENTAL-NOT-ADOPTED		6.393 ±0.035
Ca I · Codex-graded lines · 1D LTE Codex Grade · 3 lines		6.231 ±0.080
Ca I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 3 lines		6.228 ±0.073
Ca I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 3 lines		6.235 ±0.079
Ca I · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 3 lines · EXPERIMENTAL-NOT-ADOPTED		6.322 ±0.082
Ca I · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 3 lines · EXPERIMENTAL-NOT-ADOPTED		6.295 ±0.074
Ca I · reference lines · 1D LTE Reference Grade · 3 lines		6.231 ±0.080
Ca I · reference lines · 1D NLTE (Gerber) Reference Grade · 3 lines		6.228 ±0.073
Ca I · reference lines · 1D LTE (Gerber) Reference Grade · 3 lines		6.235 ±0.079
Ca I · reference lines · <3D> LTE (Gerber) Reference Grade · 3 lines · EXPERIMENTAL-NOT-ADOPTED		6.322 ±0.082
Ca I · reference lines · <3D> NLTE (Gerber) Reference Grade · 3 lines · EXPERIMENTAL-NOT-ADOPTED		6.295 ±0.074

red-optical — Kitt Peak (1984, molecfit)
A(Ca) dex; axis 6.00–6.62

Ca II · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines		6.453 ±0.052
Ca II · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 2 lines		6.466 ±0.059
Ca II · Asplund et al. 2021 lines · 1D NLTE (Bergemann) Reference Grade · 2 lines		6.417 ±0.041

red-optical — Kitt Peak (1984, molecfit)
A(Ca) dex; axis 6.00–6.62

Ca II · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 2 lines		6.318 ±0.041
Ca II · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 2 lines		6.425 ±0.049
Ca II · Asplund et al. 2021 lines · <3D> LTE (Gerber) Reference Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		6.479 ±0.051
Ca II · Asplund et al. 2021 lines · <3D> NLTE (Gerber) Reference Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		6.351 ±0.048
Ca II · Codex-graded lines · 1D LTE Codex Grade · 3 lines		6.485 ±0.042
Ca II · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 2 lines		6.466 ±0.059
Ca II · Codex-graded lines · 1D NLTE (Bergemann) Codex Grade · 3 lines		6.392 ±0.041
Ca II · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 3 lines		6.321 ±0.036
Ca II · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 3 lines		6.455 ±0.041
Ca II · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 3 lines · EXPERIMENTAL-NOT-ADOPTED		6.514 ±0.042
Ca II · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 3 lines · EXPERIMENTAL-NOT-ADOPTED		6.383 ±0.040
Ca I · Codex-graded lines · 1D LTE Codex Grade · 3 lines		6.123 ±0.091
Ca I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 3 lines		6.121 ±0.088
Ca I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 3 lines		6.127 ±0.090
Ca I · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 3 lines · EXPERIMENTAL-NOT-ADOPTED		6.209 ±0.093
Ca I · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 3 lines · EXPERIMENTAL-NOT-ADOPTED		6.176 ±0.085
Ca I · reference lines · 1D LTE Reference Grade · 3 lines		6.123 ±0.091
Ca I · reference lines · 1D NLTE (Gerber) Reference Grade · 3 lines		6.121 ±0.088

red-optical — Kitt Peak (1984, molecfit)
A(Ca) dex; axis 6.00–6.62

Ca I · reference lines · 1D LTE (Gerber) Reference Grade · 3 lines		6.127 ±0.090
Ca I · reference lines · <3D> LTE (Gerber) Reference Grade · 3 lines · EXPERIMENTAL-NOT-ADOPTED		6.209 ±0.093
Ca I · reference lines · <3D> NLTE (Gerber) Reference Grade · 3 lines · EXPERIMENTAL-NOT-ADOPTED		6.176 ±0.085

red-optical axis: 6.00 · 6.16 · 6.31 · 6.47 · 6.62 A(Ca) dex

NIR — CRIRES+ J
A(Ca) dex; axis 6.24–6.63

Ca I · Codex-graded lines · 1D LTE Codex Grade · 2 lines		6.508 ±0.040
Ca I · Codex-graded lines · 1D NLTE (Bergemann) Codex Grade · 2 lines		6.502 ±0.041
Ca I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 2 lines		6.473 ±0.042
Ca I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 2 lines		6.511 ±0.040
Ca I · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		6.562 ±0.040

NIR — Kitt Peak (1984, molecfit)
A(Ca) dex; axis 6.24–6.63

Ca I · Codex-graded lines · 1D LTE Codex Grade · 4 lines		6.439 ±0.060
Ca I · Codex-graded lines · 1D NLTE (Bergemann) Codex Grade · 4 lines		6.411 ±0.070
Ca I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 3 lines		6.396 ±0.072
Ca I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 4 lines		6.441 ±0.059
Ca I · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 4 lines · EXPERIMENTAL-NOT-ADOPTED		6.500 ±0.058
Ca I · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 3 lines · EXPERIMENTAL-NOT-ADOPTED		6.486 ±0.070

NIR axis: 6.24 · 6.34 · 6.44 · 6.53 · 6.63 A(Ca) dex

How to read this forest. Each row is one independent product, preserving its band, holding, engine, indicator and line set. Engines and indicator families are never averaged. Each bar is the uncertainty as Asplund et al. 2021 quote it: line-to-line scatter, the 3D-model term and the NLTE term (at least 0.03 dex). For

a molecule the NLTE term is replaced by its dependence on the other element, and the 3D term comes from the 3D-to-1D difference Amarsi et al. 2021 publish for the same band. Atomic-data and observational terms are not in the bar, as in Asplund; they stay in the full error budget attached to the row. Grades describe the supplied line-pool provenance, not engine quality; evaluated theoretical CNO oscillator strengths are not relabelled as laboratory evidence. Quarantined, failed and unconstrained fits have no abundance mark. Experimental products remain explicitly not adopted. Literature references provide context, never tuning targets.

How this was measured

Every calcium value on this page comes from fitting synthetic spectra to the observed solar spectrum, line by line or band by band, with the calcium abundance as the free parameter. The synthesis includes the atomic lines and the molecular opacity in every window, so blended absorption is modelled rather than attributed to the line being measured. The continuum is placed locally around every line by comparing the observation with the synthesis on its own highest pixels, and ground-based spectra are telluric-corrected across their full range.

Each indicator is measured independently on each solar spectrum and reported separately; nothing is averaged across indicators or instruments. Where a 3D or NLTE correction is published for the lines, it is applied as its own product (1D-LTE, 1D-NLTE and 3D-NLTE are separate rows). Every value carries a full uncertainty budget: fit, line strengths, microturbulence, continuum, fit window, model atmosphere, telluric residual, and the spread between spectra, each measured by refitting with that one input changed. The headline at the top is the Asplund-grade product: a published line set, the most complete treatment, then the most lines.

Band	VIS
What was measured	Ca I · Codex-graded lines; Ca II · Codex-graded lines
Solar spectra	HARPS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)
Treatments	1D-LTE, 1D-NLTE, 3D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger
Lines	2–17
Result A(Ca)	6.300–6.413 (3D-NLTE)
Largest uncertainty terms (dex)	NLTE / 3D correction 0.03, model atmosphere 0.03

Band	VIS
What was measured	Ca I · reference lines
Solar spectra	HARPS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)
Treatments	1D-LTE, 1D-NLTE, 3D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger
Lines	10–45
Result A(Ca)	6.405–6.413 (3D-NLTE)
Largest uncertainty terms (dex)	NLTE / 3D correction 0.03, model atmosphere 0.03

Band	VIS
What was measured	Ca I · Asplund et al. 2021 lines; Ca II · Asplund et al. 2021 lines
Solar spectra	HARPS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)
Treatments	1D-LTE, 1D-NLTE, 3D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger
Lines	2–10
Result A(Ca)	6.300–6.413 (3D-NLTE)

Band	VIS
What was measured	Ca I · Asplund et al. 2021 lines; Ca II · Asplund et al. 2021 lines
Largest uncertainty terms (dex)	NLTE / 3D correction 0.03, model atmosphere 0.03

Band	red-optical
What was measured	Ca I · Codex-graded lines; Ca II · Codex-graded lines
Solar spectra	IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)
Treatments	1D-LTE, 1D-NLTE, 3D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger
Lines	2–3
Result A(Ca)	6.466–6.491 (3D-NLTE)
Largest uncertainty terms (dex)	line strengths (gf) 0.07, fit / line-to-line scatter 0.05

Band	red-optical
What was measured	Ca I · reference lines
Solar spectra	IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)
Treatments	1D-LTE, 1D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger
Lines	3
Result A(Ca)	6.176–6.295 (synth-mean3D-NLTE-gerber-stagger)
Largest uncertainty terms (dex)	fit / line-to-line scatter 0.06, model atmosphere 0.03

Band	red-optical
What was measured	Ca II · Asplund et al. 2021 lines
Solar spectra	IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)
Treatments	1D-LTE, 1D-NLTE, 3D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger
Lines	2
Result A(Ca)	6.466–6.491 (3D-NLTE)
Largest uncertainty terms (dex)	line strengths (gf) 0.07, fit / line-to-line scatter 0.05

Band	NIR
What was measured	Ca I · Codex-graded lines
Solar spectra	CRIRES+ J, Kitt Peak (1984, molecfit)
Treatments	1D-LTE, 1D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger
Lines	2–4
Result A(Ca)	6.500–6.562 (synth-mean3D-LTE-gerber-stagger)
Largest uncertainty terms (dex)	line strengths (gf) 0.13, fit / line-to-line scatter 0.04

Excluded lines

These lines are left out of the Sun's measurement. They stay in the line lists: a cooler or more evolved star may measure them well. Blended lines are not excluded -- the spectrum synthesis models the blend. 5 lines.

Species	Wavelength (Å, air)	Why it is excluded
Ca I	4307.744	molecular_blend
Ca I	4434.957	atomic_blend
Ca I	4526.928	atomic_blend
Ca I	5041.618	atomic_blend
Ca I	5270.270	atomic_blend

Solar calcium context

Calcium is measured from neutral Ca I and ionized Ca II lines in the visible and red-optical. The reference lines are the ten Ca I and five Ca II lines Asplund et al. 2021 used, with the oscillator strengths they adopted; a wider set of lines with laboratory oscillator strengths and the Codex's own line selection are shown as separate products. In the infrared (Y and J bands, Kitt Peak and CRIRES+) no laboratory oscillator strength exists, so the lines use values Elgueta et al. 2024 calibrated on Cepheid stars, each with its own published uncertainty. No usable calcium line was found in the near-ultraviolet, and none in the H and K bands has an oscillator strength that was not itself tuned to the Sun.

Published line sets · gf uncertainty model · Methodology

Download the evidence

Download PDF — Solar Ca appendix

Source feed — includes quarantined fit outputs, not adopted abundances

Complete product visibility audit

Shared website/PDF report

Reproducibility manifest

Complete references and product-to-source bindings

Reproducibility

Generated from the pinned science snapshot. Source commit 798869548bcf0cc71e24af590adcc7f6c815e90e
Feed Ca.json v1.345 · updated 2026-11-07T13:11:15Z
Generated 2026-11-07T06:13:20-07:00 (source commit timestamp)
Feed SHA-256 77ccf2c0786327d3ead1d49614f53982c158881e57ac1084a7fe56760a880829
Generator scripts/generate_cno_publication.py v1.0.0

Pinned scientific source

References / Data & Model Sources

Sources are separated by their role. Registered grid lineage and quarantined diagnostics do not imply an adopted abundance or an applied correction.

See the source notes for the distinction between reference lineage and the treatment actually recorded in each feed entry.

Science and provenance source notes

Solar abundance and indicator analyses. Solar Ca comparison literature; not an adopted Codex abundance. Sources: [1], [2].

Solar abundance and indicator analyses. Published line set CA_AGSS21 (the line selection). Sources: [1].

Atomic data and gf authorities. Ca II oscillator strengths of this pool, as the product's own uncertainty budget records them (with their lineage). Sources: [1].

Model and grid sources. Feed treatment: 1D-LTE synthesis (Turbospectrum); no NLTE/3D correction claimed. Sources: [3].

Model and grid sources. Cited solar abundance differences from the published model columns, applied additively to our LTE fits; not a new in-transfer 3D calculation. Baseline atmosphere: A_marcs_lte Sources: [1].

Model and grid sources. Feed treatment: ENGINE-B-NLTE; Turbospectrum on the registered Gerber deck. The product retains its own atmosphere and ion identity. Sources: [3].

Model and grid sources. Feed treatment: synth-1D-LTE-gerber; Turbospectrum on the registered Gerber deck. The product retains its own atmosphere and ion identity. Sources: [3].

Model and grid sources. Feed treatment: synth-mean3D-LTE-gerber-stagger; Turbospectrum on the registered Gerber deck with the horizontally averaged 3D (<3D>) Stagger solar model; not a full 3D calculation. Sources: [3], [4].

Model and grid sources. Feed treatment: synth-mean3D-NLTE-gerber-stagger; Turbospectrum on the registered Gerber deck with the horizontally averaged 3D (<3D>) Stagger solar model; not a full 3D calculation. Sources: [3], [4].

Atomic data and gf authorities. Ca I oscillator strengths of this pool, as the product's own uncertainty budget records them (with their lineage). Sources: [6].

Model and grid sources. Feed treatment: 1D NLTE, per-line correction from the Amarsi 2020 GALAH departure grid via PySME (data/nlte_grids/Ca_Amarsi2020_PySME.csv). Sources: [7].

Product-to-source index

VIS · HARPS · Ca II · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3] Ca-products-dd4060b496e3

VIS · HARPS · Ca II · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [2] Ca-products-24ee92a42efb

VIS · HARPS · Ca II · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [1], [2] Ca-products-4da790d75e99

VIS · HARPS · Ca II · Asplund et al. 2021 lines · 1D NLTE (Gerber) · [1], [2], [3] Ca-products-0ea1cc06e915

VIS · HARPS · Ca II · Asplund et al. 2021 lines · 1D LTE (Gerber) · [1], [2], [3] Ca-products-3ea07dc24356

VIS · HARPS · Ca II · Asplund et al. 2021 lines · <3D> LTE (Gerber) · [1], [2], [3], [4] Ca-products-654aaa180861

VIS · HARPS · Ca II · Asplund et al. 2021 lines · <3D> NLTE (Gerber) · [1], [2], [3], [4] Ca-products-d66c461d377f

VIS · HARPS · Ca II · Codex-graded lines · 1D LTE · [1], [2], [3] Ca-products-8652ce29137d

VIS · HARPS · Ca II · Codex-graded lines · 3D NLTE (Amarsi) · [1], [2] Ca-products-161932d92295

VIS · HARPS · Ca II · Codex-graded lines · 1D NLTE (Bergemann) · [1], [2] Ca-products-96fc7a8d123f

VIS · HARPS · Ca II · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [3] Ca-products-c6c6455ce891

VIS · HARPS · Ca II · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [3] Ca-products-f3cb7b9c2618

VIS · HARPS · Ca II · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [3], [4] Ca-products-0b53689e8883

VIS · HARPS · Ca II · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [3], [4] Ca-products-3ba038572a8d

VIS · Kitt Peak (Kurucz 2005) · Ca II · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3] Ca-products-027bcd454647

VIS · Kitt Peak (Kurucz 2005) · Ca II · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [2] Ca-products-f850768f15fc

VIS · Kitt Peak (Kurucz 2005) · Ca II · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [1], [2] Ca-products-355e6142ba51

VIS · Kitt Peak (Kurucz 2005) · Ca II · Asplund et al. 2021 lines · 1D NLTE (Gerber) · [1], [2], [3] Ca-products-3eb8fe54bf53

VIS · Kitt Peak (Kurucz 2005) · Ca II · Asplund et al. 2021 lines · 1D LTE (Gerber) · [1], [2], [3] Ca-products-74fa945a500a

VIS · Kitt Peak (Kurucz 2005) · Ca II · Asplund et al. 2021 lines · <3D> LTE (Gerber) · [1], [2], [3], [4] Ca-products-61c643651ced

VIS · Kitt Peak (Kurucz 2005) · Ca II · Asplund et al. 2021 lines · <3D> NLTE (Gerber) · [1], [2], [3], [4] Ca-products-b015ac829bc9

VIS · Kitt Peak (Kurucz 2005) · Ca II · Codex-graded lines · 1D LTE · [1], [2], [3] Ca-products-32cf9f14db30

VIS · Kitt Peak (Kurucz 2005) · Ca II · Codex-graded lines · 3D NLTE (Amarsi) · [1], [2] Ca-products-96a9aa7d41a9

VIS · Kitt Peak (Kurucz 2005) · Ca II · Codex-graded lines · 1D NLTE (Bergemann) · [1], [2] Ca-products-7cdb0da60179

VIS · Kitt Peak (Kurucz 2005) · Ca II · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [3] Ca-products-350bd4845aa8

VIS · Kitt Peak (Kurucz 2005) · Ca II · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [3] Ca-products-1e22a51a371c

VIS · Kitt Peak (Kurucz 2005) · Ca II · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [3], [4] Ca-products-9ba2dcbce69d

VIS · Kitt Peak (Kurucz 2005) · Ca II · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [3], [4] Ca-products-d628f4d3b3c7

VIS · Kitt Peak (1984, molecfit) · Ca II · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3] Ca-products-b8bc6f426b97

VIS · Kitt Peak (1984, molecfit) · Ca II · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [2] Ca-products-d0e5330f7e48

VIS · Kitt Peak (1984, molecfit) · Ca II · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [1], [2] Ca-products-b6211f404e1d

VIS · Kitt Peak (1984, molecfit) · Ca II · Asplund et al. 2021 lines · 1D NLTE (Gerber) · [1], [2], [3] Ca-products-fb941e30f947

VIS · Kitt Peak (1984, molecfit) · Ca II · Asplund et al. 2021 lines · 1D LTE (Gerber) · [1], [2], [3] Ca-products-26f32c2a2c03

VIS · Kitt Peak (1984, molecfit) · Ca II · Asplund et al. 2021 lines · <3D> LTE (Gerber) · [1], [2], [3], [4] Ca-products-bf1fabae3647

VIS · Kitt Peak (1984, molecfit) · Ca II · Asplund et al. 2021 lines · <3D> NLTE (Gerber) · [1], [2], [3], [4] Ca-products-ab3c60f6dae6

VIS · Kitt Peak (1984, molecfit) · Ca II · Codex-graded lines · 1D LTE · [1], [2], [3] Ca-products-a1146ec140a6

VIS · Kitt Peak (1984, molecfit) · Ca II · Codex-graded lines · 3D NLTE (Amarsi) · [1], [2] Ca-products-21f15c91471d

VIS · Kitt Peak (1984, molecfit) · Ca II · Codex-graded lines · 1D NLTE (Bergemann) · [1], [2] Ca-products-ad06814c7d4b

VIS · Kitt Peak (1984, molecfit) · Ca II · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [3] Ca-products-c6ca4a2f3501

VIS · Kitt Peak (1984, molecfit) · Ca II · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [3] Ca-products-c4bfadb2ed65

VIS · Kitt Peak (1984, molecfit) · Ca II · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [3], [4] Ca-products-6884a1282b98

VIS · Kitt Peak (1984, molecfit) · Ca II · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [3], [4] Ca-products-037ced42dcf4

red-optical · IAG FTS · Ca II · Asplund et al. 2021 lines · 1D LTE · [1], [2], [5], [3] Ca-products-04de60a6d74b

red-optical · IAG FTS · Ca II · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [2], [5] Ca-products-16645cec819c

red-optical · IAG FTS · Ca II · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [1], [2], [5] Ca-products-da56cd5d95d5

red-optical · IAG FTS · Ca II · Asplund et al. 2021 lines · 1D NLTE (Gerber) · [1], [2], [5], [3] Ca-products-ad93557870f1

red-optical · IAG FTS · Ca II · Asplund et al. 2021 lines · 1D LTE (Gerber) · [1], [2], [5], [3] Ca-products-1fb5ed3a73e6

red-optical · IAG FTS · Ca II · Asplund et al. 2021 lines · <3D> LTE (Gerber) · [1], [2], [5], [3], [4] Ca-products-2a7960d56a98

red-optical · IAG FTS · Ca II · Asplund et al. 2021 lines · <3D> NLTE (Gerber) · [1], [2], [5], [3], [4] Ca-products-92404ec4e2f4

red-optical · IAG FTS · Ca II · Codex-graded lines · 1D LTE · [1], [2], [5], [3] Ca-products-74f520b716fd

red-optical · IAG FTS · Ca II · Codex-graded lines · 3D NLTE (Amarsi) · [1], [2], [5] Ca-products-05702e9db6b6

red-optical · IAG FTS · Ca II · Codex-graded lines · 1D NLTE (Bergemann) · [1], [2], [5] Ca-products-074deb5d51f3

red-optical · IAG FTS · Ca II · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [5], [3] Ca-products-d3000ae677d9

red-optical · IAG FTS · Ca II · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [5], [3] Ca-products-367bbf8d0737

red-optical · IAG FTS · Ca II · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [5], [3], [4] Ca-products-dcc21ec23f20

red-optical · IAG FTS · Ca II · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [5], [3], [4] Ca-products-1f7941cea5f0

red-optical · Kitt Peak (Kurucz 2005) · Ca II · Asplund et al. 2021 lines · 1D LTE · [1], [2], [5], [3] Ca-products-27dd1908bfeb

red-optical · Kitt Peak (Kurucz 2005) · Ca II · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [2], [5] Ca-products-8d29aa42818d

red-optical · Kitt Peak (Kurucz 2005) · Ca II · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [1], [2], [5] Ca-products-ebb0c89c5503

red-optical · Kitt Peak (Kurucz 2005) · Ca II · Asplund et al. 2021 lines · 1D NLTE (Gerber) · [1], [2], [5], [3] Ca-products-f88c78f715cc

red-optical · Kitt Peak (Kurucz 2005) · Ca II · Asplund et al. 2021 lines · 1D LTE (Gerber) · [1], [2], [5], [3] Ca-products-14a7716f6b95

red-optical · Kitt Peak (Kurucz 2005) · Ca II · Asplund et al. 2021 lines · <3D> LTE (Gerber) · [1], [2], [5], [3], [4] Ca-products-d69a771127ae

red-optical · Kitt Peak (Kurucz 2005) · Ca II · Asplund et al. 2021 lines · <3D> NLTE (Gerber) · [1], [2], [5], [3], [4] Ca-products-ccff2bea272b

red-optical · Kitt Peak (Kurucz 2005) · Ca II · Codex-graded lines · 1D LTE · [1], [2], [5], [3] Ca-products-56dc49239d65

red-optical · Kitt Peak (Kurucz 2005) · Ca II · Codex-graded lines · 3D NLTE (Amarsi) · [1], [2], [5] Ca-products-f25ff96b768f

red-optical · Kitt Peak (Kurucz 2005) · Ca II · Codex-graded lines · 1D NLTE (Bergemann) · [1], [2], [5] Ca-products-1db861d211ad

red-optical · Kitt Peak (Kurucz 2005) · Ca II · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [5], [3] Ca-products-eb2699d4ce05

red-optical · Kitt Peak (Kurucz 2005) · Ca II · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [5], [3] Ca-products-30829ac53e8e

red-optical · Kitt Peak (Kurucz 2005) · Ca II · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [5], [3], [4] Ca-products-0795ded0bfc3

red-optical · Kitt Peak (Kurucz 2005) · Ca II · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [5], [3], [4] Ca-products-c17e9b7efffa

red-optical · Kitt Peak (1984, molecfit) · Ca II · Asplund et al. 2021 lines · 1D LTE · [1], [2], [5], [3] Ca-products-8d803284111f

red-optical · Kitt Peak (1984, molecfit) · Ca II · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [2], [5] Ca-products-5160f873763c

red-optical · Kitt Peak (1984, molecfit) · Ca II · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [1], [2], [5] Ca-products-cc4d8695e83d

red-optical · Kitt Peak (1984, molecfit) · Ca II · Asplund et al. 2021 lines · 1D NLTE (Gerber) · [1], [2], [5], [3] Ca-products-24745645320f

red-optical · Kitt Peak (1984, molecfit) · Ca II · Asplund et al. 2021 lines · 1D LTE (Gerber) · [1], [2], [5], [3] Ca-products-64c99a90c4a9

red-optical · Kitt Peak (1984, molecfit) · Ca II · Asplund et al. 2021 lines · <3D> LTE (Gerber) · [1], [2], [5], [3], [4] Ca-products-26309cbbe2b3

red-optical · Kitt Peak (1984, molecfit) · Ca II · Asplund et al. 2021 lines · <3D> NLTE (Gerber) · [1], [2], [5], [3], [4] Ca-products-18c7973c02bd

red-optical · Kitt Peak (1984, molecfit) · Ca II · Codex-graded lines · 1D LTE · [1], [2], [5], [3] Ca-products-4d0b11a0e0f0

red-optical · Kitt Peak (1984, molecfit) · Ca II · Codex-graded lines · 3D NLTE (Amarsi) · [1], [2], [5] Ca-products-77e881e341f1

red-optical · Kitt Peak (1984, molecfit) · Ca II · Codex-graded lines · 1D NLTE (Bergemann) · [1], [2], [5] Ca-products-a73776b72c7b

red-optical · Kitt Peak (1984, molecfit) · Ca II · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [5], [3] Ca-products-725d01712945

red-optical · Kitt Peak (1984, molecfit) · Ca II · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [5], [3] Ca-products-8d3b9c5ce4e9

red-optical · Kitt Peak (1984, molecfit) · Ca II · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [5], [3], [4] Ca-products-474a613b251c

red-optical · Kitt Peak (1984, molecfit) · Ca II · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [5], [3], [4] Ca-products-fdc9c0fc4ffc

NIR · CRIRES+ J · Ca I · Codex-graded lines · 1D LTE · [1], [2], [6], [3] Ca-products-25f0f6318518

NIR · CRIRES+ J · Ca I · Codex-graded lines · 1D NLTE (Bergemann) · [1], [2], [6], [7] Ca-products-cf629c836f94

NIR · CRIRES+ J · Ca I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [6], [3] Ca-products-15bab7aefe14

NIR · CRIRES+ J · Ca I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [6], [3] Ca-products-65390a11fbe5

NIR · CRIRES+ J · Ca I · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [6], [3], [4] Ca-products-9cf6cde28a0e

NIR · CRIRES+ J · Ca I · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [6], [3], [4] Ca-products-ef9d2a954a57

NIR · Kitt Peak (1984, molecfit) · Ca I · Codex-graded lines · 1D LTE · [1], [2], [6], [3] Ca-products-70fae1d82110

NIR · Kitt Peak (1984, molecfit) · Ca I · Codex-graded lines · 1D NLTE (Bergemann) · [1], [2], [6], [7] Ca-products-dd18535d790c

NIR · Kitt Peak (1984, molecfit) · Ca I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [6], [3] Ca-products-c59c55eb834e

NIR · Kitt Peak (1984, molecfit) · Ca I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [6], [3] Ca-products-ebbe28d99e34

NIR · Kitt Peak (1984, molecfit) · Ca I · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [6], [3], [4] Ca-products-97bbbb607baf

NIR · Kitt Peak (1984, molecfit) · Ca I · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [6], [3], [4] Ca-products-2b3bfe45a534

VIS · HARPS · Ca I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [8], [3] Ca-products-67afa4d144c4

VIS · HARPS · Ca I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [2], [8] Ca-products-594066726043

VIS · HARPS · Ca I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [1], [2], [8] Ca-products-2e4d1bfdedeb

VIS · HARPS · Ca I · Asplund et al. 2021 lines · 1D NLTE (Gerber) · [1], [2], [8], [3] Ca-products-1681e5bc05e9

VIS · HARPS · Ca I · Asplund et al. 2021 lines · 1D LTE (Gerber) · [1], [2], [8], [3] Ca-products-942c4706548b

VIS · HARPS · Ca I · Asplund et al. 2021 lines · <3D> LTE (Gerber) · [1], [2], [8], [3], [4] Ca-products-f6069426e3db

VIS · HARPS · Ca I · Asplund et al. 2021 lines · <3D> NLTE (Gerber) · [1], [2], [8], [3], [4] Ca-products-bd502817bc63

VIS · HARPS · Ca I · Codex-graded lines · 1D LTE · [1], [2], [8], [3] Ca-products-870e697b4548

VIS · HARPS · Ca I · Codex-graded lines · 3D NLTE (Amarsi) · [1], [2], [8] Ca-products-8b605ffca287

VIS · HARPS · Ca I · Codex-graded lines · 1D NLTE (Bergemann) · [1], [2], [8] Ca-products-16bfde2a9b53

VIS · HARPS · Ca I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [8], [3] Ca-products-6242d03efa4d

VIS · HARPS · Ca I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [8], [3] Ca-products-0117886eb350

VIS · HARPS · Ca I · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [8], [3], [4] Ca-products-ca1ad271ee45

VIS · HARPS · Ca I · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [8], [3], [4] Ca-products-14d65377d4d2

VIS · HARPS · Ca I · reference lines · 1D LTE · [1], [2], [8], [3] Ca-products-938a8ab1cc61

VIS · HARPS · Ca I · reference lines · 3D NLTE (Amarsi) · [1], [2], [8] Ca-products-770124cd45c0

VIS · HARPS · Ca I · reference lines · 1D NLTE (Bergemann) · [1], [2], [8] Ca-products-2a3a5dcbc544

VIS · HARPS · Ca I · reference lines · 1D NLTE (Gerber) · [1], [2], [8], [3] Ca-products-f32339fdad5b

VIS · HARPS · Ca I · reference lines · 1D LTE (Gerber) · [1], [2], [8], [3] Ca-products-e97cde189ab8

VIS · HARPS · Ca I · reference lines · <3D> LTE (Gerber) · [1], [2], [8], [3], [4] Ca-products-da098b831ac9

VIS · HARPS · Ca I · reference lines · <3D> NLTE (Gerber) · [1], [2], [8], [3], [4] Ca-products-f39300bb0faa

VIS · Kitt Peak (Kurucz 2005) · Ca I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [8], [3] Ca-products-77ff4dc80d76

VIS · Kitt Peak (Kurucz 2005) · Ca I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [2], [8] Ca-products-f6b767bc1fdb

VIS · Kitt Peak (Kurucz 2005) · Ca I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [1], [2], [8] Ca-products-aaeed3a89804

VIS · Kitt Peak (Kurucz 2005) · Ca I · Asplund et al. 2021 lines · 1D NLTE (Gerber) · [1], [2], [8], [3] Ca-products-0aca6a28dbd5

VIS · Kitt Peak (Kurucz 2005) · Ca I · Asplund et al. 2021 lines · 1D LTE (Gerber) · [1], [2], [8], [3] Ca-products-2f0225028134

VIS · Kitt Peak (Kurucz 2005) · Ca I · Asplund et al. 2021 lines · <3D> LTE (Gerber) · [1], [2], [8], [3], [4] Ca-products-501866f6674c

VIS · Kitt Peak (Kurucz 2005) · Ca I · Asplund et al. 2021 lines · <3D> NLTE (Gerber) · [1], [2], [8], [3], [4] Ca-products-e5a78a4d43d9

VIS · Kitt Peak (Kurucz 2005) · Ca I · Codex-graded lines · 1D LTE · [1], [2], [8], [3] Ca-products-b748feac0a21

VIS · Kitt Peak (Kurucz 2005) · Ca I · Codex-graded lines · 3D NLTE (Amarsi) · [1], [2], [8] Ca-products-040b43d2304f

VIS · Kitt Peak (Kurucz 2005) · Ca I · Codex-graded lines · 1D NLTE (Bergemann) · [1], [2], [8] Ca-products-6b395e49f892

VIS · Kitt Peak (Kurucz 2005) · Ca I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [8], [3] Ca-products-0ec8ff786657

VIS · Kitt Peak (Kurucz 2005) · Ca I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [8], [3] Ca-products-7978d7087995

VIS · Kitt Peak (Kurucz 2005) · Ca I · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [8], [3], [4] Ca-products-69cd236e6085

VIS · Kitt Peak (Kurucz 2005) · Ca I · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [8], [3], [4] Ca-products-170c815a288e

VIS · Kitt Peak (Kurucz 2005) · Ca I · reference lines · 1D LTE · [1], [2], [8], [3] Ca-products-4b281ba118ad

VIS · Kitt Peak (Kurucz 2005) · Ca I · reference lines · 3D NLTE (Amarsi) · [1], [2], [8] Ca-products-4a61217108a0

VIS · Kitt Peak (Kurucz 2005) · Ca I · reference lines · 1D NLTE (Bergemann) · [1], [2], [8] Ca-products-361188dae0c0

VIS · Kitt Peak (Kurucz 2005) · Ca I · reference lines · 1D NLTE (Gerber) · [1], [2], [8], [3] Ca-products-3de49ca51fb9

VIS · Kitt Peak (Kurucz 2005) · Ca I · reference lines · 1D LTE (Gerber) · [1], [2], [8], [3] Ca-products-c3c5cd7b0f7f

VIS · Kitt Peak (Kurucz 2005) · Ca I · reference lines · <3D> LTE (Gerber) · [1], [2], [8], [3], [4] Ca-products-6bd593347fe1

VIS · Kitt Peak (Kurucz 2005) · Ca I · reference lines · <3D> NLTE (Gerber) · [1], [2], [8], [3], [4] Ca-products-e98cf6a7f2db

VIS · Kitt Peak (1984, molecfit) · Ca I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [8], [3] Ca-products-516dfde46042

VIS · Kitt Peak (1984, molecfit) · Ca I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [2], [8] Ca-products-527c8f910d64

VIS · Kitt Peak (1984, molecfit) · Ca I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [1], [2], [8] Ca-products-077029de6067

VIS · Kitt Peak (1984, molecfit) · Ca I · Asplund et al. 2021 lines · 1D NLTE (Gerber) · [1], [2], [8], [3] Ca-products-ec9781239520

VIS · Kitt Peak (1984, molecfit) · Ca I · Asplund et al. 2021 lines · 1D LTE (Gerber) · [1], [2], [8], [3] Ca-products-971099823d57

VIS · Kitt Peak (1984, molecfit) · Ca I · Asplund et al. 2021 lines · <3D> LTE (Gerber) · [1], [2], [8], [3], [4] Ca-products-a2b6df7692f1

VIS · Kitt Peak (1984, molecfit) · Ca I · Asplund et al. 2021 lines · <3D> NLTE (Gerber) · [1], [2], [8], [3], [4] Ca-products-958fa01618b2

VIS · Kitt Peak (1984, molecfit) · Ca I · Codex-graded lines · 1D LTE · [1], [2], [8], [3] Ca-products-deecc7e22814

VIS · Kitt Peak (1984, molecfit) · Ca I · Codex-graded lines · 3D NLTE (Amarsi) · [1], [2], [8] Ca-products-e7f3d4442690

VIS · Kitt Peak (1984, molecfit) · Ca I · Codex-graded lines · 1D NLTE (Bergemann) · [1], [2], [8] Ca-products-1e6a7f516aa8

VIS · Kitt Peak (1984, molecfit) · Ca I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [8], [3] Ca-products-83f0d7cddd02

VIS · Kitt Peak (1984, molecfit) · Ca I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [8], [3] Ca-products-2a1ee3c2ca4a

VIS · Kitt Peak (1984, molecfit) · Ca I · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [8], [3], [4] Ca-products-bdfcfd167157

VIS · Kitt Peak (1984, molecfit) · Ca I · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [8], [3], [4] Ca-products-d75157d93e65

VIS · Kitt Peak (1984, molecfit) · Ca I · reference lines · 1D LTE · [1], [2], [8], [3] Ca-products-4db7a5da0003

VIS · Kitt Peak (1984, molecfit) · Ca I · reference lines · 3D NLTE (Amarsi) · [1], [2], [8] Ca-products-cdeb56c453aa

VIS · Kitt Peak (1984, molecfit) · Ca I · reference lines · 1D NLTE (Bergemann) · [1], [2], [8] Ca-products-0a1a25d9e5a1

VIS · Kitt Peak (1984, molecfit) · Ca I · reference lines · 1D NLTE (Gerber) · [1], [2], [8], [3] Ca-products-784fc345ba03

VIS · Kitt Peak (1984, molecfit) · Ca I · reference lines · 1D LTE (Gerber) · [1], [2], [8], [3] Ca-products-1f2dfa604246

VIS · Kitt Peak (1984, molecfit) · Ca I · reference lines · <3D> LTE (Gerber) · [1], [2], [8], [3], [4] Ca-products-e80b9e23221b

VIS · Kitt Peak (1984, molecfit) · Ca I · reference lines · <3D> NLTE (Gerber) · [1], [2], [8], [3], [4] Ca-products-906bb9b24ef3

red-optical · IAG FTS · Ca I · Codex-graded lines · 1D LTE · [1], [2], [8], [3] Ca-products-840d08e05a54

red-optical · IAG FTS · Ca I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [8], [3] Ca-products-8b359e9693d2

red-optical · IAG FTS · Ca I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [8], [3] Ca-products-c94cf539bdae

red-optical · IAG FTS · Ca I · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [8], [3], [4] Ca-products-18e5f1433a4b

red-optical · IAG FTS · Ca I · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [8], [3], [4] Ca-products-f2874fcf7406

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